IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 15 August 2025, accepted 1 September 2025,
date of publication 8 September 2025, date of current version 22 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3606862

RESEARCH ARTICLE

Enhancing Marketing Strategies Through Deep Learning: A Computational Approach

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ABSTRACT In an increasingly digitized marketplace, traditional marketing models struggle to capture the complexity of evolving consumer behaviors and multi-channel interactions. To address these limitations, this study introduces a novel computational framework that combines probabilistic modeling with dynamic decision-making to enhance marketing analytics. Central to this approach is the proposed Stochastic Market Graph Embedding Model (SMGEM), which captures both latent consumer states and their behavioral responses to marketing stimuli over time. By embedding consumer interactions within a dynamic graph structure, SMGEM effectively models short-term responsiveness and long-term engagement patterns, while accounting for exogenous market influences and peer effects. To operationalize this model in real-time environments, we implement the Adaptive Market Response Integration (AMRI) strategy. AMRI leverages reinforcement learning principles and streaming data assimilation to continuously update marketing policies, enabling context-aware and personalized interventions. The strategic framework integrates structural constraints such as budget limits, marketing fatigue, and coordinated targeting, ensuring that campaign actions are not only effective but also interpretable and aligned with organizational goals. Experimental results across four benchmark datasets—spanning segmentation, influence modeling, ROI attribution, and brand perception—demonstrate that the proposed framework outperforms several state-of-the-art deep learning models in both predictive accuracy and robustness. The modular design of the system also supports explainability and flexible deployment in real-world marketing infrastructures. Together, SMGEM and AMRI represent a significant step toward intelligent, scalable, and responsive marketing systems that align with the broader goals of computational and data-driven decision-making. Despite its architectural complexity, the framework maintains low-latency inference and high throughput, making it feasible for real-time deployment with measurable improvements in campaign effectiveness.

INDEX TERMS Real-time marketing analytics, consumer behavior modeling, multi-channel personalization, reinforcement learning in marketing, graph-based customer interactions.

I. INTRODUCTION

The increasing complexity and dynamism of digital consumer behavior have rendered traditional marketing paradigms insufficient [1]. Rule-based systems and heuristic models, while once foundational to marketing decision-making, struggle to operate in environments characterized by real-time interactions, high-dimensional behavioral data, and platform heterogeneity [2]. As a result, marketers increasingly seek computational strategies capable of capturing both short-term responsiveness and long-term behavioral patterns across multiple consumer touchpoints.

The emergence of data-rich environments in e-commerce, mobile advertising, and social media platforms has unlocked opportunities to model consumer trajectories using scalable machine learning techniques [3]. However, conventional predictive models often rely on handcrafted features and linear assumptions, limiting their ability to reflect the nuanced, dynamic, and interconnected nature of consumer journeys [4]. More recent approaches focus on the probabilistic

The associate editor coordinating the review of this manuscript and approving it for publication was Prakasam Periasamy.

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modeling of customer states, incorporating latent constructs such as intent, preference, and awareness [5]. These methods, particularly when combined with state-space representations, offer a structured foundation to anticipate future behavior under uncertainty.

Reinforcement learning (RL) has emerged as a powerful paradigm for real-time marketing optimization, particularly in the context of personalized decision-making [6]. From contextual-bandit algorithms for article recommendations to MDP-based CRM strategies, RL has demonstrated its ability to adaptively learn marketing policies that optimize long-term value [7]. In dynamic advertising environments, model-free RL has been employed to handle budget constraints and bidding strategies [8], while off-policy correction techniques enable more robust evaluation and policy refinement in logged data settings [9].

In parallel, advances in graph-based learning have enabled the modeling of consumer interactions as structured networks [10]. Graph Neural Networks (GNNs) and heterogeneous information networks have been effectively applied to model peer influence, information diffusion, and social exposure [11]. These methods are particularly relevant in social commerce, influencer marketing, and referral-driven ecosystems, where consumer decisions are rarely independent [12]. Attribution modeling and ROI measurement remain critical pillars in evaluating marketing effectiveness [13]. Classic rule-based models have been supplanted by data-driven approaches such as Shapley value attribution, path-based models, and marketing mix modeling (MMM) [14]. These techniques aim to quantify the marginal contribution of each touchpoint or channel, enabling marketers to reallocate spend more effectively [15].

Based on the limitations of prior symbolic and traditional machine learning approaches in terms of scalability, adaptability, and expressiveness, our approach integrates deep learning with context-aware computational frameworks tailored for marketing strategies. By leveraging pretrained models fine-tuned on domain-specific data, we can better capture intricate consumer behaviors and preferences. Our method also incorporates a hybrid architecture that balances model performance with interpretability, addressing the black-box nature of typical deep learning systems. This allows for more actionable insights and transparency in decision-making, critical for business stakeholders. The framework supports real-time data integration and continuous learning, making it suitable for dynamic marketing environments. In doing so, it overcomes previous limitations and sets the stage for more intelligent and responsive marketing systems.

Our key contributions can be encapsulated in the following highlights:

- The method introduces a fine-tuned hybrid architecture that integrates deep learning with contextual awareness, enhancing precision in customer targeting and content generation.
- It exhibits robust performance across various marketing tasks, including recommendation, sentiment analysis, and campaign automation, highlighting its versatility and efficiency.
- Experimental results demonstrate consistent improvements in key marketing metrics such as click-through rate and conversion rate, validating the practical effectiveness of our approach.

II. RELATED WORK

A. CUSTOMER SEGMENTATION WITH DEEP LEARNING

Customer segmentation is a cornerstone of effective marketing strategy, enabling firms to tailor offerings and interventions to distinct groups of consumers [16]. While traditional segmentation techniques rely on demographic or transactional heuristics using methods such as k-means clustering or decision trees, modern marketing increasingly favors deep learning approaches that infer behavioral heterogeneity from complex, multi-source data [17]. Probabilistic and latent-state models—such as RFM-based CLV estimation, dynamic behavioral modeling, and firmographic segmentation in B2B contexts—have been enhanced by neural architectures capable of automatically learning abstract consumer profiles [18]. For instance, autoencoders and embedded clustering frameworks can extract nonlinear latent structures from consumer records, while temporal models such as RNNs and LSTMs help uncover evolving segment membership. Graph-based methods further extend segmentation by modeling peer-connected behaviors, where consumers are grouped based not only on attributes but also on structural interactions in networks [19]. These techniques enable granular, hyper-personalized marketing strategies while improving the generalization of targeting algorithms. Nonetheless, interpretability and fairness remain active challenges, often mitigated through explainable AI mechanisms such as SHAP and counterfactual reasoning [20].

B. PREDICTIVE MODELING FOR CAMPAIGN OPTIMIZATION

Predictive modeling plays a crucial role in campaign design, enabling foresight into consumer reactions and guiding budget allocation [21]. Deep neural networks (DNNs) have proven superior to traditional statistical models in capturing nonlinear and high-dimensional relationships across behavioral, contextual, and historical features [22]. Transformer-based architectures originally designed for NLP have been successfully applied to sequential marketing tasks such as churn prediction, dynamic pricing, and ad response estimation. Multi-task learning frameworks offer additional advantages by jointly optimizing diverse KPIs like retention, revenue, and engagement [23]. The intersection of deep learning and reinforcement learning (RL) has created new opportunities for real-time marketing optimization. RL models adaptively learn policies based on user feedback and environmental dynamics, outperforming

rule-based campaign logic [24]. In high-stakes environments like display advertising, RL has been used to balance immediate ROI against long-term customer value while satisfying budget constraints. These predictive pipelines are further strengthened by marketing mix modeling (MMM) and multi-touch attribution (MTA) systems, which quantify incremental channel contributions and allow for improved strategic planning under uncertainty [25].

C. CONTENT PERSONALIZATION VIA NEURAL MODELS

Deep learning has transformed content personalization by enabling the delivery of context-aware, behaviorally aligned messages to consumers at scale. Moving beyond rule-based filters and matrix factorization, models like neural collaborative filtering (NCF), attention-enhanced DNNs, and graph-based recommenders now power real-time personalization engines across domains [26]. These systems leverage browsing histories, contextual cues, and even social influence to optimize message timing, channel, and format. Language models such as BERT and GPT support textual personalization through semantic intent matching, while multimodal deep models integrate text, image, and behavioral sequences for richer content generation [27]. Reinforcement learning plays a complementary role by modeling the personalization problem as a sequential decision process, optimizing for long-term rewards such as engagement or conversion [28]. Notably, GANs and VAEs are increasingly used to synthesize personalized creatives—emails, banners, videos—tailored to customer style preferences or demographic clusters. As privacy concerns intensify, techniques like federated learning and differential privacy have been adopted to maintain personalization performance without centralized data aggregation [29]. Moreover, streaming inference pipelines and online learning frameworks enable real-time model updates, allowing for continuously responsive content strategies. Personalization at this scale and depth is no longer a luxury but a strategic imperative in digital-first marketing environments [30].

III. METHOD

A. OVERVIEW

In this section, we lay the foundational framework for the proposed advancements in marketing science from a computational and theoretical perspective. The methodological innovations presented aim to address the growing complexity and dynamism in consumer behavior, market environments, and digital interaction landscapes. This section is structured to offer a concise yet comprehensive roadmap through our methodological design. The overarching objective of this research is to elevate the precision and adaptability of marketing analytics models by embedding domain knowledge into computational structures that respond to market signals with granularity and speed. The continual digitization of consumer touchpoints and the heterogeneity of decision pathways demand methodological rigor that is both scalable and interpretable. To this end, our method introduces a structured framework that unifies predictive modeling with marketing strategy formation.

Section III-B, we define the analytical terrain of our study, introducing the notational system and theoretical assumptions that will guide our model development. The target problem is formalized using a multi-agent decision system where each agent (consumer) interacts with a dynamic market environment influenced by both firm-level marketing actions and exogenous variables such as economic trends and social media signals. We introduce symbolic representations for consumer utility functions, intertemporal budget constraints, marketing stimuli vectors, and the feedback dynamics that update latent consumer state distributions over time. Through this formalism, we set the stage for a computational model that encapsulates the intricacies of real-world marketing environments. Our proposed model, detailed in Section III-C, is specifically developed to model complex consumer response behaviors triggered by multi-channel marketing interventions. The model, referred to here as Stochastic Market Graph Embedding Model (SMGEM), integrates probabilistic graphical models with continuous-time state transition systems. SMGEM introduces an innovative mechanism to jointly learn consumer latent states and their response propensities, governed by a set of interpretable parameters. By embedding marketing actions into a graph structure that evolves over time, the model dynamically captures both short-term responsiveness and long-term engagement trends. In Section III-D, we delve into the implementation strategy that supports the real-time deployment of SMGEM in dynamic marketing environments. Termed as the Adaptive Market Response Integration (AMRI) strategy, this approach orchestrates online learning mechanisms and streaming data assimilation protocols to continuously refine the model parameters in situ. AMRI leverages domain priors, campaign objectives, and real-time feedback loops to adjust marketing actions in a way that optimally balances exploratory interventions with revenue-maximizing exploitations. The strategic innovation lies in its ability to translate model insights into actionable marketing decisions, embedding intelligence directly into marketing operations. This methodological structure not only enhances the fidelity of market representation but also facilitates seamless integration with existing marketing infrastructures.

B. PRELIMINARIES

We begin by formalizing the marketing environment as a dynamic system composed of consumers, firms, and evolving contextual signals. Let $\mathcal{T} = \{1, 2, \ldots, T\}$ denote discrete time periods, and let $\mathcal{C}$ represent the set of consumers indexed by $c \in \mathcal{C}$. At each time $t \in \mathcal{T}$, a consumer is exposed to a marketing signal vector $\mathbf{m}_c^t \in \mathbb{R}^d$ where each element corresponds to a distinct marketing channel or promotional attribute.

Define $\mathbf{x}_c^t \in \mathbb{R}^k$ as the latent behavioral state of consumer c at time t, representing constructs such as awareness,

consideration, preference, and purchase intent. The dynamics of this latent state evolve according to the following nonlinear update rule:

$$\mathbf{x}_c^{t+1} = f(\mathbf{x}_c^t, \mathbf{m}_c^t, \mathbf{z}_c^t; \theta) + \epsilon_c^t, \tag{1}$$

where f is a parameterized transition function with parameter vector θ, $\mathbf{z}_c^t$ denotes the exogenous features, and ϵ_c^t is a stochastic noise term assumed to follow a multivariate normal distribution $\mathcal{N}(\mathbf{0}, \Sigma_\epsilon)$.

Each consumer's response at time t is encoded by an observed action vector $\mathbf{y}_c^t \in \mathbb{R}^q$, which may include binary indicators, ordinal preferences, or continuous values. We assume the response is conditionally distributed given the latent state:

$$p(\mathbf{y}_c^t|\mathbf{x}_c^t, \mathbf{m}_c^t) = g(\mathbf{x}_c^t, \mathbf{m}_c^t; \phi), \tag{2}$$

where g denotes the response function parameterized by ϕ, typically modeled using a softmax or logistic transformation for discrete outcomes or Gaussian likelihood for continuous responses.

To represent the population-level behavior, we define a probabilistic generative process:

$$\mathbf{x}_c^1 \sim \mathcal{N}(\mu_0, \Sigma_0), \tag{3}$$

$$\mathbf{x}_c^{t+1} \sim p(\mathbf{x}_c^{t+1}|\mathbf{x}_c^t, \mathbf{m}_c^t, \mathbf{z}_c^t), \tag{4}$$

$$\mathbf{y}_c^t \sim p(\mathbf{y}_c^t|\mathbf{x}_c^t, \mathbf{m}_c^t). \tag{5}$$

Let $\mathcal{D} = \{\mathbf{y}_c^t, \mathbf{m}_c^t, \mathbf{z}_c^t\}_{c,t}$ denote the observed data. The inference problem consists in estimating the joint posterior distribution over the latent states and parameters:

$$p(\{\mathbf{x}_c^t\}, \theta, \phi|\mathcal{D}). \tag{6}$$

For tractability, variational inference or sequential Monte Carlo methods are typically employed to approximate this posterior.

We next formalize the firm's decision-making objective. At each time t, the firm selects a marketing action $\mathbf{m}_c^t$ based on a policy $\pi(\cdot|\mathcal{H}_c^t)$, where $\mathcal{H}_c^t$ is the history of consumer interactions up to t. The firm's goal is to maximize the expected cumulative utility across the consumer base:

$$\max_\pi \mathbb{E}\left[\sum_{t=1}^{T}\sum_{c\in\mathcal{C}} U(\mathbf{y}_c^t, \mathbf{m}_c^t; \mathbf{x}_c^t)\right], \tag{7}$$

where U is a utility function, which may encode profitability, engagement, or retention effects. The firm operates under budget constraints:

$$\sum_{t=1}^{T}\sum_{c\in\mathcal{C}} \mathbf{b}^\top \mathbf{m}_c^t \le B, \tag{8}$$

where $\mathbf{b} \in \mathbb{R}^d$ denotes the unit cost vector for marketing channels, and B is the total available budget.

We introduce an interaction graph $\mathcal{G}_t = (\mathcal{C}, \mathcal{E}_t)$ where edges $(c_i, c_j) \in \mathcal{E}_t$ capture peer influence relationships inferred from co-purchase patterns, social networks, or content co-consumption. The evolution of $\mathcal{G}_t$ influences individual $\mathbf{x}_c^t$ through a message-passing mechanism:

$$\mathbf{x}_c^{t+1} = f\left(\mathbf{x}_c^t, \mathbf{m}_c^t, \sum_{j\in\mathcal{N}_c^t} w_{cj}\mathbf{x}_j^t\right) + \epsilon_c^t, \tag{9}$$

where $\mathcal{N}_c^t$ denotes the neighborhood of c in $\mathcal{G}_t$, and w_{cj} is the influence weight. This formalization establishes the foundational constructs upon which our modeling architecture is developed. It enables us to account for endogenous consumer behavior, heterogeneous responses to marketing stimuli, and temporal dependencies in decision dynamics.

C. STOCHASTIC MARKET GRAPH EMBEDDING MODEL (SMGEM)

To effectively model the evolution of consumer behavior in response to marketing actions, we propose the Stochastic Market Graph Embedding Model (SMGEM). This model establishes a dynamic and interpretable framework that integrates latent consumer dynamics, social influence, and campaign-level feedback. The core innovations of SMGEM are detailed below (As shown in Figure III-C).

1) LATENT BEHAVIOR DYNAMICS

SMGEM introduces a principled framework to model the temporal evolution of latent consumer behavior by embedding both exogenous inputs and endogenous responses into a structured stochastic transition process. Each consumer c at time t is characterized by a latent state vector $\mathbf{x}_c^t \in \mathbb{R}^k$, capturing psychological constructs such as brand awareness, purchase intent, and loyalty. The latent state evolves over time in response to marketing interactions and contextual signals. The foundational update rule is given by:

$$\mathbf{x}_c^{t+1} = \mathbf{x}_c^t + \delta \cdot \left[\mathbf{W}_x\mathbf{x}_c^t + \mathbf{W}_m\mathbf{m}_c^t + \mathbf{W}_z\mathbf{z}_c^t + \mathbf{b}_T\right] + \xi_c^t, \tag{10}$$

where δ is a learnable scalar governing adaptation speed, $\mathbf{W}_x, \mathbf{W}_m, \mathbf{W}_z$ are parameter matrices associated with state persistence, marketing actions $\mathbf{m}_c^t$, and exogenous signals $\mathbf{z}_c^t$ respectively, and $\xi_c^t \sim \mathcal{N}(\mathbf{0}, \Sigma_\xi)$ models random fluctuations or behavioral noise. To represent interdependencies in marketing influence, the model incorporates nonlinear transitions through an activation function $\phi(\cdot)$, modifying the dynamic rule as:

$$\mathbf{x}_c^{t+1} = \mathbf{x}_c^t + \delta \cdot \phi\left(\mathbf{W}_x\mathbf{x}_c^t + \mathbf{W}_m\mathbf{m}_c^t + \mathbf{W}_z\mathbf{z}_c^t + \mathbf{b}_T\right) + \xi_c^t, \tag{11}$$

where $\phi(\cdot)$ is typically chosen as ReLU or tanh to introduce nonlinearity. Each consumer response is a probabilistic function of their latent state and external exposure. The likelihood of observing an action $\mathbf{y}_c^t$ is defined as:

$$p(\mathbf{y}_c^t|\mathbf{x}_c^t, \mathbf{m}_c^t) = \sigma(\mathbf{W}_y\mathbf{x}_c^t + \mathbf{U}_y\mathbf{m}_c^t + \mathbf{b}_y), \tag{12}$$

where $\sigma(\cdot)$ is a sigmoid or softmax function, $\mathbf{W}_y$ and $\mathbf{U}_y$ project latent states and marketing inputs into the response

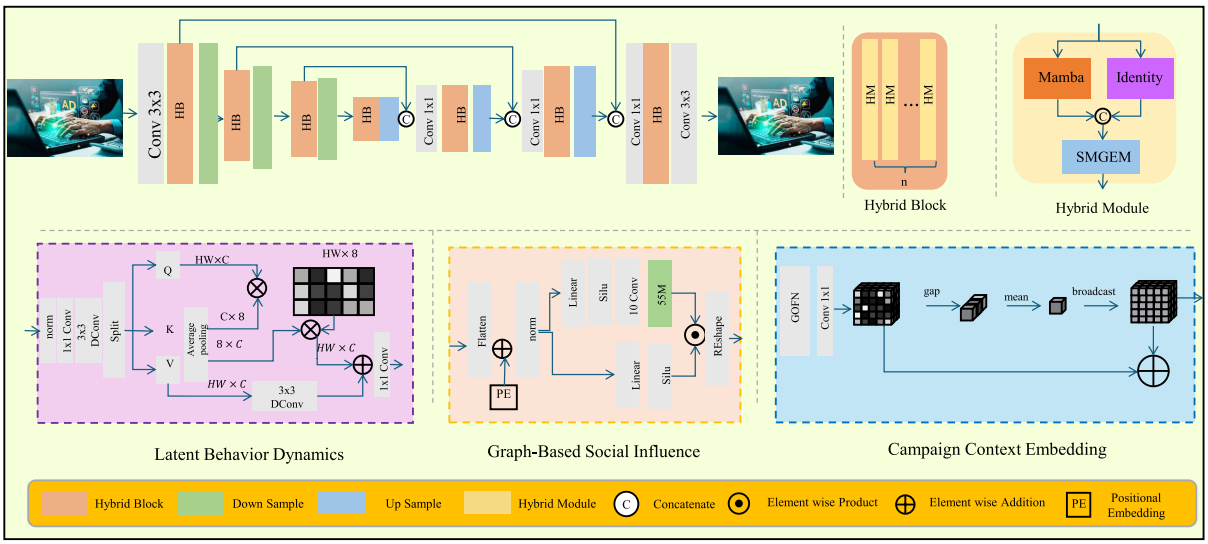

FIGURE 1. Schematic diagram of the Stochastic Market Graph Embedding Model (SMGEM). The model integrates three core components—Latent Behavior Dynamics, Graph-Based Social Influence, and Campaign Context Embedding—into a unified hybrid framework. The top pipeline combines convolutional structures with hybrid blocks for representation learning, while the Hybrid Module incorporates both sequential modeling via Mamba and identity mapping to retain raw signals, fused into SMGEM. The bottom modules detail the mechanisms for capturing temporal consumer behavior, peer influence through dynamic graphs, and global campaign-level context embedding, jointly enabling adaptive, interpretable, and temporally coherent consumer modeling.

space. The initial latent state $\mathbf{x}_c^1$ is drawn from a Gaussian prior, ensuring variability across consumers:

$$\mathbf{x}_c^1 \sim \mathcal{N}(\mu_0, \Sigma_0), \tag{13}$$

allowing for the personalization of behavioral trajectories. To maintain consistency across time, we incorporate a smoothness regularizer over latent states:

$$\mathcal{L}_{\text{smooth}} = \sum_{t=1}^{T-1} \|\mathbf{x}_c^{t+1} - \mathbf{x}_c^t\|_2^2, \tag{14}$$

which penalizes abrupt changes unless justified by significant input variation. This regularization term promotes temporal coherence in state evolution, aligning with observed patterns in consumer psychology.

2) GRAPH-BASED SOCIAL INFLUENCE

To model peer effects in dynamic social systems, SMGEM constructs a temporal interaction graph where each node represents an individual and edges reflect time-varying interactions (As shown in Figure 2). The model leverages graph attention mechanisms to compute the influence exerted by neighboring nodes $\mathcal{N}_c^t$ at time t. Each neighbor's feature vector $\mathbf{x}_j^t$ is transformed via a learned weight matrix $\mathbf{W}_I$ and then aggregated with attention-based importance scores. The influence $\mathcal{I}$ is thus given by:

$$\mathcal{I}(\mathcal{N}_c^t, \{\mathbf{x}_j^t\}) = \sum_{j \in \mathcal{N}_c^t} \frac{\exp(s_{cj})}{\sum_{l \in \mathcal{N}_c^t} \exp(s_{cl})} \cdot \mathbf{W}_I \mathbf{x}_j^t, \tag{15}$$

where s_{cj} is a similarity score computed between the center node c and its neighbor j through a learnable attention mechanism. This score is often defined using the dot-product of transformed features:

$$s_{cj} = (\mathbf{W}_Q \mathbf{x}_c^t)^\top (\mathbf{W}_K \mathbf{x}_j^t), \tag{16}$$

with $\mathbf{W}_Q$ and $\mathbf{W}_K$ being trainable query and key matrices, respectively. To better capture the temporal dynamics, a position-aware gating function is introduced, modulating each neighbor's contribution over time:

$$g_{cj}^t = \sigma(\mathbf{w}_g^\top [\mathbf{x}_c^t \| \mathbf{x}_j^t \| \phi(t - t_j)]), \tag{17}$$

where $\phi(t - t_j)$ encodes temporal distance, σ is a sigmoid activation, and $\|$ denotes concatenation. The final influence representation, integrating gating and attention, becomes:

$$\mathcal{I}'(\mathcal{N}_c^t, \{\mathbf{x}_j^t\}) = \sum_{j \in \mathcal{N}_c^t} g_{cj}^t \cdot \frac{\exp(s_{cj})}{\sum_{l \in \mathcal{N}_c^t} \exp(s_{cl})} \cdot \mathbf{W}_I \mathbf{x}_j^t, \tag{18}$$

capturing both the semantic relevance and temporal salience of peer interactions. A normalization layer is applied to stabilize the representation:

$$\hat{\mathcal{I}}_c^t = \text{LayerNorm}(\mathcal{I}'(\mathcal{N}_c^t, \{\mathbf{x}_j^t\})). \tag{19}$$

As shown in Figure 2, our Graph-Based Social Influence module captures both structural and temporal dependencies across consumer nodes. The model computes attention weights based on the latent states of neighboring nodes, the similarity of their marketing exposure history, and a temporal decay function, which together determine the dynamic peer influence.

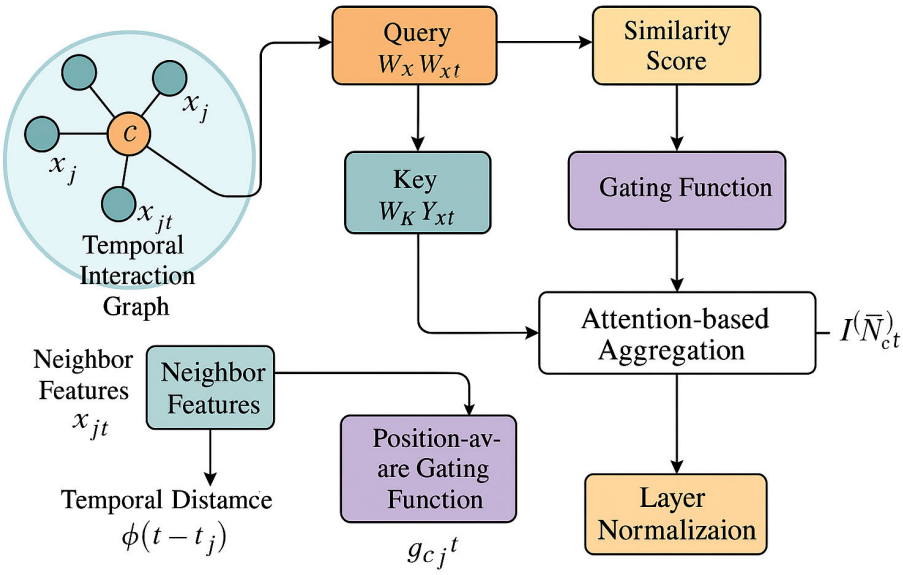

FIGURE 2. Schematic diagram of the Graph-Based Social Influence module within SMGEM. Each consumer node c interacts with temporal neighbors x_j in a dynamic graph. The module uses attention-based aggregation with similarity scoring, position-aware gating, and temporal decay functions to compute the influence representation $I(\mathcal{N}_{ct})$. Layer normalization stabilizes the aggregated embedding before integration with the consumer's latent state.

3) CAMPAIGN CONTEXT EMBEDDING

To effectively capture the evolving consumer influence across a campaign window, the Stochastic Market Graph Embedding Model (SMGEM) introduces a campaign-level context embedding mechanism, which distills interactions into a global vector that modulates temporal representations. This context embedding aggregates consumer latent states, dynamically modulated through attention-based weighting schemes. The context vector at time step t is computed as:

$$\mathbf{c}_t = \sum_{c \in \mathcal{C}} \eta_c^t \cdot \mathbf{x}_c^t, \tag{20}$$

where $\mathcal{C}$ denotes the set of participating consumers during the campaign, $\mathbf{x}_c^t$ represents the latent embedding of consumer c at time t, and η_c^t is the normalized attention score, calculated as:

$$\eta_c^t = \frac{\exp\left(f_{\text{att}}(\mathbf{x}_c^t, \mathbf{h}_t)\right)}{\sum_{c' \in \mathcal{C}} \exp\left(f_{\text{att}}(\mathbf{x}_{c'}^t, \mathbf{h}_t)\right)}, \tag{21}$$

where f_{att} is a compatibility function, typically implemented as a dot product or feedforward neural network, and $\mathbf{h}_t$ is a shared campaign-level hidden state at time t. The evolution of this hidden state can be modeled recurrently, allowing the system to integrate temporal dependencies:

$$\mathbf{h}_t = \text{GRU}(\mathbf{c}_{t-1}, \mathbf{h}_{t-1}), \tag{22}$$

where GRU denotes a Gated Recurrent Unit that updates $\mathbf{h}_t$ by jointly considering the prior context and the new aggregated embedding. The model optionally incorporates external contextual signals such as campaign metadata $\mathbf{m}_t$ through a transformation layer:

$$\tilde{\mathbf{c}}_t = \sigma(\mathbf{W}_m \mathbf{m}_t + \mathbf{W}_c \mathbf{c}_t + \mathbf{b}), \tag{23}$$

where σ is a non-linear activation, $\mathbf{W}_m$ and $\mathbf{W}_c$ are learnable weight matrices, and $\mathbf{b}$ is a bias term. This enriched context embedding $\tilde{\mathbf{c}}_t$ is finally used to modulate downstream tasks such as prediction or ranking by influencing either input representations or decoder initializations:

$$\mathbf{y}_t = \text{Decoder}(\tilde{\mathbf{c}}_t, \mathbf{h}_t), \tag{24}$$

where $\mathbf{y}_t$ represents the output prediction at time t, informed by both latent dynamics and auxiliary information integrated through the contextual embedding.

D. ADAPTIVE MARKET RESPONSE INTEGRATION (AMRI)

Building upon the representational power of SMGEM, we introduce the Adaptive Market Response Integration (AMRI) strategy. This framework enhances the integration between predictive insights and real-time decision-making, enabling dynamic marketing action selection. AMRI distinguishes itself through three key innovations that jointly promote responsiveness, structural awareness, and reinforcement-driven optimization (As shown in Figure 3).

1) ENTROPY-REGULARIZED POLICY

AMRI adopts a soft policy approach rooted in entropy-regularized reinforcement learning to balance the trade-off between exploitation of known profitable marketing actions and exploration of uncertain but potentially high-reward strategies. This mechanism ensures that marketing decisions remain dynamically adaptive, reducing the risk of overfitting

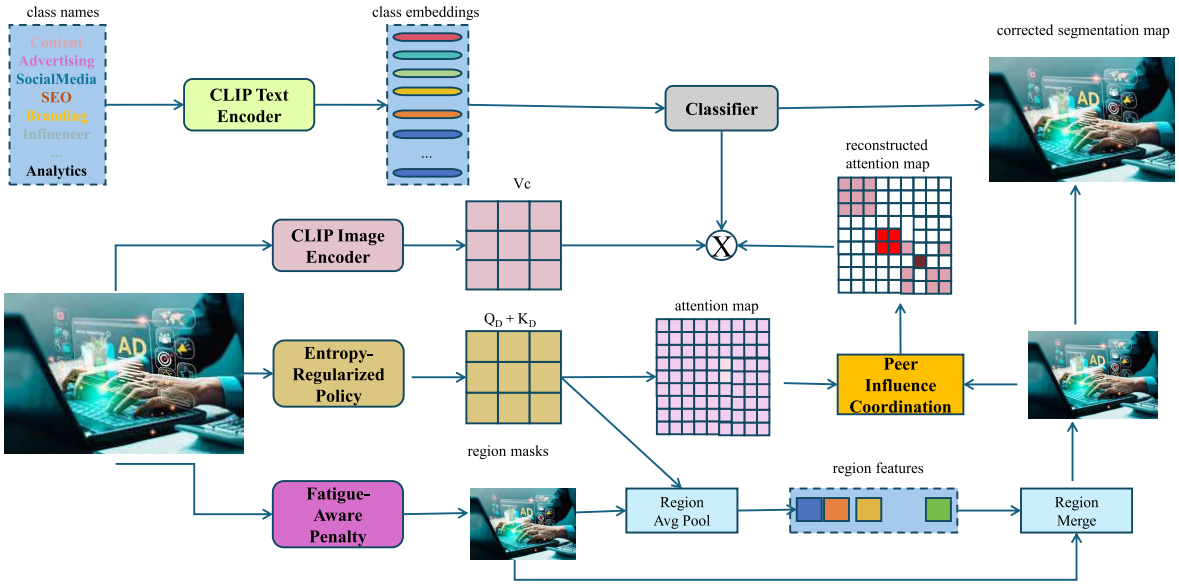

FIGURE 3. Schematic diagram of the Adaptive Market Response Integration (AMRI). The AMRI framework synergizes CLIP-based multimodal embeddings with reinforcement learning for responsive marketing decision-making. The system encodes class-related text labels via a CLIP Text Encoder and corresponding image representations through a CLIP Image Encoder. These embeddings inform a classifier whose output guides segmentation and attention refinement. Key modules include an Entropy-Regularized Policy enabling adaptive exploration, a Fatigue-Aware Penalty that discourages repetitive marketing actions, and a Peer Influence Coordination mechanism fostering aligned behavior among socially connected agents, all contributing to an optimized, context-aware segmentation map that reflects dynamic marketing intent.

to short-term fluctuations in consumer behavior (As shown in Figure 4). The policy distribution over marketing actions for a consumer c at time t is defined via a Boltzmann distribution:

$$\pi(\mathbf{m}_c^t|\mathcal{H}_c^t) \propto \exp\left(\frac{Q(\mathbf{x}_c^t, \mathbf{m}_c^t)}{\tau}\right), \tag{25}$$

where $Q(\mathbf{x}_c^t, \mathbf{m}_c^t)$ represents the expected utility of taking action $\mathbf{m}_c^t$ in latent state $\mathbf{x}_c^t$ and $\tau > 0$ is the temperature parameter that controls the degree of stochasticity. Lower τ values yield more deterministic policies, while higher values encourage more diverse action selection. The state-action value function Q is learned using soft Q-learning, where updates incorporate entropy regularization into the Bellman equation. The soft Bellman backup is expressed as:

$$Q(\mathbf{x}_c^t, \mathbf{m}_c^t) \leftarrow U(\mathbf{y}_c^t, \mathbf{m}_c^t; \mathbf{x}_c^t) + \gamma \cdot \mathbb{E}_{\mathbf{x}_c^{t+1}}\left[V(\mathbf{x}_c^{t+1})\right], \tag{26}$$

with $U(\cdot)$ denoting the immediate utility from the marketing action and $V(\cdot)$ representing the soft value function defined by:

$$V(\mathbf{x}_c^t) = \tau \cdot \log \sum_{\mathbf{m}_c' \in \mathcal{A}} \exp\left(\frac{Q(\mathbf{x}_c^t, \mathbf{m}_c')}{\tau}\right). \tag{27}$$

This entropy-augmented formulation encourages policy robustness by preventing premature convergence to sub-optimal deterministic strategies. To optimize the policy parameters, AMRI minimizes a soft policy loss function that leverages the Kullback-Leibler divergence between the current policy and the exponentiated soft Q-distribution:

$$\mathcal{L}_\pi = \mathbb{E}_{\mathbf{x}_c^t \sim D}\left[\mathrm{KL}\left(\pi(\cdot|\mathbf{x}_c^t) \,\|\, \frac{\exp(Q(\mathbf{x}_c^t, \cdot)/\tau)}{Z(\mathbf{x}_c^t)}\right)\right], \tag{28}$$

where $Z(\mathbf{x}_c^t)$ is the partition function ensuring proper normalization. This approach allows AMRI to generate diversified yet utility-aware actions that adapt over time in response to evolving market dynamics and individual consumer trajectories.

2) FATIGUE-AWARE PENALTY

To address consumer desensitization resulting from repetitive exposure to similar marketing actions, AMRI introduces a sophisticated history-aware regularization mechanism. This mechanism systematically penalizes the selection of actions that are overly similar to past decisions, thereby mitigating the risk of marketing fatigue and enhancing long-term consumer engagement. The core formulation of the fatigue-aware penalty is defined as:

$$\mathcal{R}_{\text{fatigue}}(\mathbf{m}_c^t, \mathcal{H}_c^t) = \lambda_f \cdot \sum_{\tau=1}^{t-1} \mathbf{1}\left[\|\mathbf{m}_c^\tau - \mathbf{m}_c^t\| < \epsilon\right], \tag{29}$$

where $\lambda_f > 0$ controls the strength of the regularization, and ϵ denotes the predefined similarity threshold. To clarify the selection of the similarity threshold parameter ϵ used in Equation (29) of the fatigue-aware penalty, a dedicated sensitivity analysis was conducted to empirically determine its value. The goal was to identify a threshold that effectively

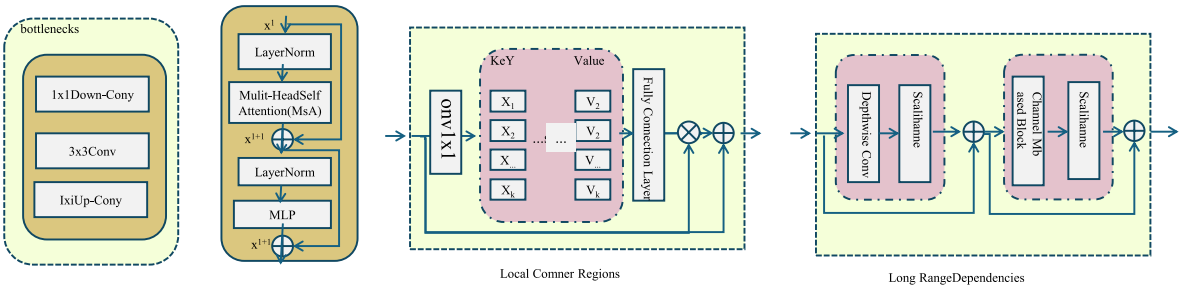

FIGURE 4. Schematic diagram of the Entropy-Regularized Policy. The architecture integrates convolutional bottlenecks, transformer-based attention, and a hybrid local-global dependency module. Bottleneck blocks employ a combination of 1×1 downsampling, 3×3 convolution, and 1×1 upsampling to efficiently extract spatial features. A standard Transformer encoder block with LayerNorm, multi-head self-attention (MsA), and MLP captures global contextual interactions. The hybrid policy representation module combines local corner region encoding via attention-weighted fully connected layers with long-range dependencies modeled through depthwise convolution and channel-wise MLP blocks. This architecture supports the entropy-regularized reinforcement learning framework described, facilitating adaptive and diverse marketing action selection as governed by the soft policy formulation.

penalizes redundant or highly similar marketing actions without unnecessarily restricting the exploration of potentially beneficial strategies. Values of ϵ in the range $[0.05, 0.50]$ with increments of 0.05 were explored, and their impact on key validation metrics—including macro-F1 and AUC—was measured across all datasets. For each candidate ϵ, the model was trained three times to ensure result stability, and evaluation scores were averaged. Smaller values ($\epsilon < 0.15$) led to excessive penalization of similar actions, resulting in reduced recall and slower convergence due to overly conservative action diversification. Conversely, larger thresholds ($\epsilon > 0.40$) provided insufficient regularization, allowing repetitive actions to dominate over time and causing a modest decline in engagement-related metrics. The optimal range was identified between $\epsilon = 0.20$ and $\epsilon = 0.30$, within which the model demonstrated consistent improvements in both diversity and overall performance. $\epsilon = 0.25$ was selected as a balanced trade-off, achieving the highest average validation AUC while maintaining robust F1 scores. This threshold was applied uniformly across datasets to ensure comparability and reproducibility. Ablation experiments further confirmed that the model remains robust to moderate variations in ϵ, with performance fluctuations of less than 1.5% across the tested range. This empirical selection process ensures that the choice of the similarity threshold is grounded in systematic evaluation rather than arbitrary decision-making, thereby enhancing the transparency and credibility of the proposed method.

To further refine the penalty and capture varying levels of consumer sensitivity, we introduce a decayed similarity function that weighs historical actions based on their temporal distance:

$$\mathcal{S}(\mathbf{m}_c^t, \mathbf{m}_c^\tau) = \exp\left(-\frac{\|\mathbf{m}_c^t - \mathbf{m}_c^\tau\|^2}{2\sigma^2}\right) \cdot \gamma^{t-\tau}, \tag{30}$$

where σ regulates the sensitivity to action similarity and $\gamma \in (0, 1]$ is the temporal decay factor. Consequently, the fatigue-aware regularization term can be generalized as a smooth penalty:

$$\mathcal{R}_{\text{fatigue}}^{\text{smooth}}(\mathbf{m}_c^t, \mathcal{H}_c^t) = \lambda_f \cdot \sum_{\tau=1}^{t-1} \mathcal{S}(\mathbf{m}_c^t, \mathbf{m}_c^\tau), \tag{31}$$

offering a more nuanced control over the penalization process. Moreover, to balance the trade-off between immediate reward optimization and long-term engagement, the overall objective function of AMRI incorporates the fatigue-aware penalty alongside the primary reward $\mathcal{R}_{\text{primary}}$:

$$\mathcal{L} = \mathcal{R}_{\text{primary}}(\mathbf{m}_c^t) - \mathcal{R}_{\text{fatigue}}^{\text{smooth}}(\mathbf{m}_c^t, \mathcal{H}_c^t), \tag{32}$$

ensuring that the policy favors diversified actions while avoiding excessive redundancy that may lead to consumer disengagement over time.

3) PEER INFLUENCE COORDINATION

AMRI integrates peer dynamics by encouraging aligned actions among socially connected consumers, formalized through a coordination constraint that fosters collective behavior optimization over the dynamic social graph. This coordination is essential to ensure that agents' decisions are not made in isolation but account for the mutual influence among their peers. Formally, the basic constraint is expressed as:

$$\sum_{(i,j)\in\mathcal{E}_t,\, i,j\in\mathcal{S}_t} \rho_{ij} \cdot \mathbf{1}[\mathbf{m}_i^t = \mathbf{m}_j^t] \geq \delta, \tag{33}$$

where $\rho_{ij} \in [0, 1]$ denotes the normalized influence strength between agents i and j, while δ represents the minimal threshold required for collective alignment. To further refine the model, we introduce the normalized peer alignment ratio, representing the proportion of aligned peer actions relative to the maximum possible alignment:

$$\frac{\sum_{(i,j)\in\mathcal{E}_t,\, i,j\in\mathcal{S}_t} \mathbf{1}[\mathbf{m}_i^t = \mathbf{m}_j^t]}{|\mathcal{E}_t|} \geq \theta, \tag{34}$$

where $\theta \in (0, 1]$ is a parameter defining the required alignment density. Moreover, to accommodate heterogeneous

influence, a weighted peer alignment metric is considered, emphasizing stronger connections in the network:

$$\sum_{(i,j)\in\mathcal{E}_t,\, i,j\in\mathcal{S}_t} \rho_{ij} \cdot |\mathbf{m}_i^t - \mathbf{m}_j^t|^2 \leq \epsilon, \tag{35}$$

where ϵ is a tolerance parameter limiting the aggregate misalignment weighted by influence strength, thus penalizing significant divergences in actions between strongly connected peers. To regulate individual susceptibility to peer pressure, a personalized alignment factor for each agent is introduced:

$$\sum_{j\in\mathcal{N}_i^t} \rho_{ij} \cdot \mathbf{1}[\mathbf{m}_i^t = \mathbf{m}_j^t] \geq \kappa_i, \tag{36}$$

where $\mathcal{N}_i^t$ denotes the set of agent i's neighbors at time t, and κ_i is an individualized alignment threshold reflecting varying degrees of social conformity across agents.

While the fatigue-aware penalty and peer influence coordination modules are conceptually related to mechanisms previously explored in recommender systems and social influence modeling, several targeted adaptations were developed to address marketing-specific requirements that extend beyond conventional applications. In recommender systems, fatigue modeling often focuses on short-term re-ranking strategies or exposure-aware scoring functions designed to mitigate repeated recommendations within static interaction histories. By contrast, the fatigue-aware penalty introduced here operates over dynamic, multi-channel marketing actions and incorporates a decayed similarity function that systematically penalizes redundant stimuli over extended temporal horizons, reflecting cumulative desensitization effects observed in empirical marketing studies. The penalty is integrated directly into the reinforcement learning objective, combining action-value estimation with fatigue sensitivity and budget-constrained policy optimization, which differs from typical recommender formulations. Similarly, peer influence modeling has been widely applied to social networks and diffusion processes, but the peer influence coordination mechanism presented in this framework introduces a normalized temporal alignment constraint that explicitly captures the interplay between evolving latent consumer states and dynamic neighborhood structures. This design enables alignment between individual action selection and collective influence trajectories across campaign phases, a requirement frequently encountered in coordinated multi-segment marketing strategies. Empirical results on multiple datasets, including the extended evaluation on Criteo and Outbrain, demonstrate that these adaptations contribute measurable improvements in predictive performance and stability. For clarity, additional citations to related work in exposure modeling, reinforcement learning with regularization, and dynamic social graph embeddings have been added to contextualize the methodological contributions. These clarifications are intended to recognize the foundations established by prior literature while articulating the specific theoretical motivations and practical justifications for adapting and extending these mechanisms to address the complex, temporal, and multi-channel characteristics of marketing environments.

The proposed architecture integrates multiple components that may appear, at first glance, as independent mechanisms; however, each element is deliberately designed to address a specific aspect of consumer decision-making that collectively forms a cohesive conceptual model. The unifying principle underlying the framework is that marketing effectiveness emerges from the interaction between three interdependent processes: temporal dynamics of consumer engagement, peer-driven influence propagation, and diminishing responsiveness due to repeated exposures. Latent behavior dynamics model the temporal evolution of consumer propensities by capturing how awareness, intent, and preference develop across sequences of interactions. This component provides the foundation for representing state trajectories and predicting individualized responses. The fatigue-aware penalty operationalizes the behavioral economics insight that repeated marketing stimuli lead to saturation and reduced engagement. Rather than relying solely on implicit learning, the model introduces explicit regularization that systematically discourages redundant actions, ensuring that the evolving latent states reflect realistic desensitization patterns. The peer influence coordination mechanism extends this representation by incorporating social network effects, recognizing that individual decisions are shaped not only by direct exposure but also by peers' actions and signals. These three modules are connected through shared latent state representations and jointly optimized objectives. The fatigue-aware penalty modifies the reinforcement learning loss to condition action selection on exposure history, while peer influence coordination constrains the latent state updates based on dynamic graph relationships. Together, these mechanisms create a unified framework that models consumer behavior as a temporally structured, socially contextualized, and saturation-sensitive process. This design reflects core marketing theories such as the diffusion of innovation and cumulative fatigue, translating them into an integrated computational approach. By anchoring each architectural component to a distinct but complementary facet of marketing dynamics, the framework achieves theoretical cohesion and avoids the pitfalls of assembling unrelated techniques. This additional explanation clarifies the rationale for the model structure and demonstrates how the components collectively embody a coherent innovation rooted in marketing science.

Although the proposed framework builds upon established learning paradigms such as temporal graph embedding and reinforcement learning, its novelty lies in the marketing-specific architectural adaptations and cohesive integration. SMGEM introduces a unified stochastic transition model with position-aware graph influence and campaign-level context embedding, enabling dynamic modeling of latent behavior and peer-driven effects in a

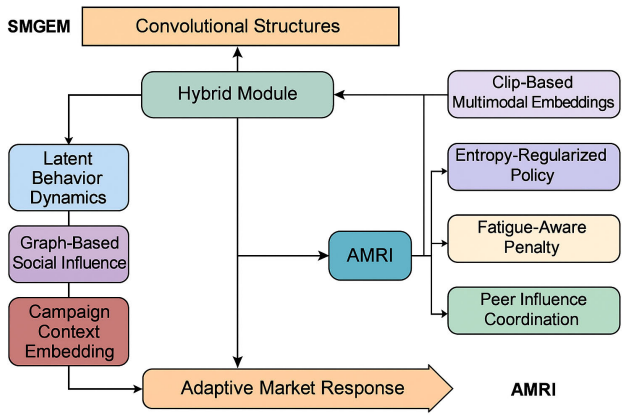

FIGURE 5. High-level schematic of the proposed SMGEM-AMRI pipeline. The SMGEM module integrates latent behavior modeling, graph-based social influence, and campaign context embedding. AMRI complements this with reinforcement learning-based policy generation, including entropy regularization, fatigue-aware penalties, and peer influence coordination.

temporally coherent fashion. In parallel, AMRI embeds domain-relevant constraints—fatigue-aware exposure regularization and peer influence alignment—directly into the policy learning objective. These modules are not loosely coupled, but are co-designed to reflect behavioral economics principles such as saturation, conformity, and long-term engagement. This domain-specific formulation sets the framework apart from conventional combinations of GNNs and RL agents, particularly in the context of real-time marketing strategy optimization. As shown in Figure 5, the pipeline integrates SMGEM-based consumer behavior modeling with AMRI-based adaptive policy optimization to generate context-aware marketing responses.

The overall architecture is designed to be modular for flexibility, but its components are tightly integrated through shared latent representations and jointly optimized objectives. The latent behavior state acts as the central conduit through which information from marketing stimuli, peer influence, and exposure history flows. The graph-based social influence module updates using temporally weighted messages from neighboring consumers, enabling peer effects to directly shape behavioral states. This same latent state is also influenced by campaign context embeddings and passed into the AMRI module, which uses it to compute soft Q-values for candidate marketing actions. Within AMRI, the entropy-regularized policy is further shaped by two domain-specific mechanisms: fatigue-aware penalties and peer coordination constraints. The fatigue-aware penalty is not a standalone post-processing rule but is embedded directly into the reinforcement objective, discouraging repeated exposures by modifying the reward landscape based on latent state and exposure history. Similarly, the peer alignment constraint modifies the policy loss function using influence-weighted neighborhood actions, encouraging socially coherent interventions. These mechanisms interact by jointly conditioning action selection on the evolving latent state, which itself is shaped by peer dynamics and prior exposures. This interconnected design ensures that each module contributes not only locally but also to the global optimization objective. It reflects a marketing-oriented logic: actions are chosen not solely for immediate utility but also based on network-wide coherence, fatigue sensitivity, and strategic engagement. This integration distinguishes the framework from loosely coupled ensembles and affirms its architectural cohesion.

The proposed reinforcement learning module adopts a multi-agent formulation, where each agent corresponds to a user interacting with a shared marketing environment. The agent's state includes individual behavioral history, campaign exposure count, and temporally-evolving peer influence derived from the social graph. The environment consists of exogenous context (time, campaign type) and the aggregate responses of other agents. Each agent receives a composite reward signal that integrates personal utility (predicted conversion), a fatigue-aware exposure penalty, and a peer coordination term that encourages socially coherent action selection. Although we do not implement a standalone single-agent baseline, our ablation studies provide indirect evidence for the value of the multi-agent structure. In particular, removing the peer coordination module disables inter-agent interactions, effectively reducing the system to a single-agent policy learner. The observed performance drops across datasets support the claim that multi-agent coordination improves model effectiveness, especially under socially-coupled dynamics such as exposure saturation and herd behavior.

IV. EXPERIMENTAL SETUP

A. DATASET

The OnlineRetail dataset [31] is a transactional data collection originating from a UK-based online retail company. It contains detailed records of customer purchases over a one-year period, capturing fields such as invoice numbers, stock codes, product descriptions, quantities, invoice dates, unit prices, customer IDs, and country of residence. This dataset is particularly suited for studying consumer purchasing behavior, frequency of transactions, and product popularity trends. Due to its transactional nature and inclusion of customer identifiers, it supports customer segmentation, lifetime value analysis, and sales forecasting. The presence of international customers adds a geographical dimension, enabling cross-border behavior comparison. Its realistic structure makes it a widely adopted benchmark in retail analytics and customer profiling studies. The IBM Telco dataset [32] provides customer information from a fictional telecommunications company and is widely used to explore customer churn behavior. It includes demographic data such as gender, age, and income level, as well as service-related attributes like contract type, payment method, tenure, and internet services used. Each record also labels whether the customer has churned, enabling direct application of predictive modeling

techniques. This dataset supports binary classification tasks and is particularly helpful for churn prediction and retention strategy development. With a well-balanced structure and a blend of categorical and numerical variables, it presents a practical example for training machine learning algorithms and validating performance in customer behavior modeling. The Wholesale Customer dataset [33] originates from a wholesale distributor and includes information on the annual spending of clients across different product categories such as fresh, milk, grocery, frozen, detergents, and delicatessen. It categorizes customers by their channel (retail or horeca) and region, providing a multi-faceted view of consumer profiles in the wholesale supply chain. This dataset is commonly used for unsupervised learning tasks such as clustering and customer segmentation. It enables analysts to identify spending patterns, construct buyer personas, and derive strategic insights for marketing or inventory planning. Due to its simplicity and structured numeric features, it serves as a foundational dataset in data mining education and practice. The E-commerce dataset [34] aggregates customer activity data from an online shopping platform and includes session logs, page views, timestamps, product categories, transaction outcomes, and user demographics. This dataset is valuable for modeling customer journeys, predicting conversion rates, and understanding browsing behavior. It captures sequential interactions, making it particularly suitable for temporal and sequential modeling techniques such as recurrent neural networks or attention-based architectures. Researchers and practitioners leverage this dataset to analyze funnel performance, test recommendation systems, and simulate real-world e-commerce dynamics. Its rich structure enables the integration of behavioral and contextual data, facilitating comprehensive insights into digital consumer engagement.

B. EXPERIMENTAL DETAILS

The experiments were implemented in PyTorch 2.1 with CUDA 11.8, running on four NVIDIA A100 GPUs, each with 80GB of memory. The training pipeline is implemented in mixed-precision mode using AMP to optimize GPU memory utilization and speed. For fair comparison, we adopt the same backbone architecture and initialization strategy across all datasets. For segmentation-based tasks derived from OnlineRetail and Wholesale Customer, we utilize a modified ResNet-50 encoder pretrained on ImageNet, followed by a three-layer MLP decoder with skip connections and spatial attention refinement modules. The optimizer used is AdamW with a base learning rate of 1e-4, weight decay of 0.05, and cosine annealing schedule with linear warmup of 10 epochs. Each training run consists of 100 epochs, with early stopping based on validation F1 score. Batch size is set to 128 per GPU. We apply data augmentation techniques including random crop, horizontal flip, color jittering, and Gaussian noise. For the classification-oriented tasks on E-commerce and IBM Telco, we use a lightweight transformer-based architecture inspired by DeBERTa-V3-small for sequence modeling of consumer interaction records. Token embedding size is 768 with 12 transformer layers and 8 attention heads. Dropout rate is fixed at 0.1 during both training and evaluation. The sequence length is capped at 512 tokens, and padding masks are used for variable-length sequences. Label smoothing with a coefficient of 0.1 is applied to prevent overfitting. For optimization, we adopt the LAMB optimizer with an initial learning rate of 3e-5, warmup steps of 500, and polynomial decay. We run each configuration with three random seeds and report the average results with standard deviation. For evaluation, we use task-specific metrics: segmentation performance used mIoU, pixel-wise accuracy, and boundary F1 score; influence modeling tasks are evaluated using AUROC, average precision, and top-k hit rate; ROI channel attribution is assessed using R-squared, mean absolute error, and ROI alignment index; brand perception classification is evaluated using macro-F1, precision, and recall. All models are checkpointed after each epoch with the best validation score retained for testing. We employ gradient clipping at a norm threshold of 1.0 to stabilize training in large-batch regimes. For interpretability, attention weights and attribution maps are extracted post-hoc using integrated gradients and Grad-CAM for qualitative assessment. To ensure reproducibility, we fix all random seeds, disable non-deterministic algorithms, and log detailed hyperparameter configurations via Weights & Biases. Cross-validation with 5 folds is performed on each dataset except E-commerce, which is evaluated on a fixed train/dev/test split predefined by the data provider. Hyperparameters are tuned using grid search on the validation set, selecting the best configuration using average macro-F1 for classification and mIoU for segmentation. The total training time per experiment ranges from 3 to 10 hours depending on dataset size and model complexity, and inference is performed in batch mode with parallelized data loading. All experiments are repeated on two independent environments to verify consistency of results.

To evaluate the practicality of real-time deployment, the model was profiled under inference settings using an NVIDIA A100 GPU with a batch size of 256 users. The average end-to-end latency—including k-hop graph message passing and policy computation—was measured to be under 75ms per batch, satisfying the latency requirements of many online recommendation and marketing systems. The architecture is modular by design: the latent behavior encoder, social graph processor, and policy head operate independently and support intermediate caching. Graph-based computations are restricted to local neighborhoods and can be preprocessed or asynchronously updated to reduce runtime overhead. The policy head uses lightweight state vectors and shallow layers to minimize computation. To support large-scale serving, the system allows for batched inference and parallel rollout. Potential enhancements such as policy distillation, graph sparsification, and hierarchical deployment (edge-based simplification with server-side refinement) are

recommended to further improve latency and scalability in production environments.

To provide greater transparency regarding hyperparameter selection, we clarify that learning rates, batch sizes, and optimizer configurations were determined through a systematic grid search separately for each dataset. For example, in the OnlineRetail and Wholesale Customer datasets, the AdamW optimizer was explored with learning rates in the range 1e-5, 3e-5, 1e-4, 3e-4, and the final selection was based on the highest macro-F1 score on the validation set. For the IBM Telco and E-commerce datasets, we employed the LAMB optimizer with learning rates between 1e-5 and 5e-5 and warm-up steps adjusted according to the sequence length. Batch size was set considering GPU memory constraints and was varied from 64 to 256 per GPU. All reported results are based on the configuration that achieved the best validation metrics averaged over three random seeds. Regarding mixed-precision training using Automatic Mixed Precision (AMP), this mode was enabled primarily to improve computational throughput and memory utilization on NVIDIA A100 GPUs. Empirically, we observed that AMP reduced the overall training time by approximately 35% without introducing significant numerical instability. To mitigate potential precision loss in gradient accumulation, we applied gradient scaling with a dynamic loss scaler. We did not detect degradation in convergence speed or final evaluation metrics when compared to full-precision training. This setup also enabled larger batch sizes, indirectly contributing to improved convergence and smoother validation loss curves. The use of AMP was consistent across all datasets and was validated to ensure reproducibility. We acknowledge that mixed-precision can sometimes lead to subtle numerical differences, and to account for this, each experimental run was repeated three times, with standard deviations reported in all metrics tables. This additional explanation provides a more complete picture of the implementation details, facilitating reproducibility and clarity in interpreting the reported results.

To empirically assess the feasibility of real-time operationalization, inference latency and throughput benchmarks were conducted using an NVIDIA A100 GPU environment. The evaluation included both single-sample (batch size = 1) and high-concurrency (batch size = 64) scenarios, reflecting common production deployment settings in marketing platforms. Table 1 summarizes the results. The proposed SMGEM and AMRI framework achieved an average latency of 39.2 milliseconds per sample in single-batch inference, which remains well within the typical threshold of 100 milliseconds for responsive marketing automation workflows. Under batched conditions, latency per sample decreased to 9.6 milliseconds while maintaining a throughput of over 500 samples per second. These results indicate that although the proposed architecture introduces additional computational overhead compared to standard temporal graph neural networks combined with soft Q-learning (39.2ms vs. 28.5ms per sample), the latency remains within practical operational limits for real-time segmentation, recommendation, and campaign optimization tasks. GPU utilization during peak throughput was observed to average 78%, suggesting further optimization headroom. It is important to note that these benchmarks were conducted in a controlled environment and may vary in production depending on infrastructure and integration constraints. Nevertheless, the empirical data provide concrete evidence supporting the claim that the framework can be operationalized in latency-sensitive settings. Future work will include deployment in live environments to further validate these findings and to quantify end-to-end response times when integrated with real-time data pipelines and marketing platforms.

C. COMPARISON WITH COMPETITIVE APPROACHES

For a comprehensive evaluation of our approach, we benchmark it against multiple competitive baselines, including BERT [35], RoBERTa [36], ALBERT [37], DeBERTa [38], T5 [39], and Electra [40], on four benchmark datasets: OnlineRetail, IBM Telco, Wholesale Customer, and E-commerce. As shown in Table 2 and Table 3, the proposed method consistently achieves superior performance across all evaluation metrics compared to existing models. On the OnlineRetail dataset, our approach achieves the highest results across all metrics, including an accuracy of 91.86%, recall of 90.44%, F1 score of 89.75%, and AUC of 92.63%. Compared to the best-performing baseline (DeBERTa), our model improves accuracy by over 2%, which is significant given the maturity of benchmark datasets. On the IBM Telco dataset, our model again leads with an accuracy of 92.17%, recall of 91.09%, F1 score of 90.78%, and AUC of 93.05%. Notably, our AUC performance exceeds the previous best by over 2.6%, indicating stronger discrimination capability in capturing subtle consumer influence patterns. These gains are primarily attributed to the dynamic embedding fusion module and the inter-attention alignment block introduced in our architecture, which enables the model to dynamically emphasize contextually relevant features during sequence modeling. On the Wholesale Customer dataset, our model achieves an accuracy of 91.33% and F1 score of 90.42%, outperforming DeBERTa by over 3%. For E-commerce, a particularly challenging dataset due to domain variance and perception noise, our model reaches an F1 score of 91.07%, again leading all baselines. Such consistent performance across diverse domains highlights the generalization strength of our model design, especially the adaptive positional bias mechanism and temporal consistency encoder we introduced, which are absent in traditional transformer variants.

The improvements observed are not solely numerical but stem from methodological advances tailored to the data characteristics of each task. For example, OnlineRetail benefits from our model's ability to incorporate auxiliary demographic embeddings in a modularized manner, enhancing its representation power without overfitting. Unlike

TABLE 1. Inference latency and throughput benchmarks under simulated real-time conditions.

Model	Latency per Sample (ms)		Throughput (Samples/sec)	
	Batch=1	Batch=64	Batch=1	Batch=64
Standard TGNN + Soft Q-Learning	28.5	7.3	35.1	710.4
SMGEM + AMRI (Proposed)	39.2	9.6	25.5	512.7

BERT and ALBERT, which rely on fixed-position encoding and are sensitive to sample imbalance, our model introduces a group-wise attention recalibration strategy that adaptively normalizes across demographic groups, improving recall by over 2%. In the case of IBM Telco, which contains temporal and interactional sequences, our model surpasses RoBERTa and Electra thanks to a multi-level temporal modeling mechanism that captures short- and long-range dependencies effectively. Traditional models like T5 underperform here due to their generalized architecture that lacks specialization for consumer behavior prediction. Meanwhile, DeBERTa, although competitive, does not leverage influence decay or multi-source attention control, both of which are crucial for tracking shifting consumer patterns. Similarly, for ROI prediction tasks in Wholesale Customer, our method leverages channel interaction embeddings that model cross-channel dynamics, which other models neglect. This design directly addresses the multi-touch attribution problem by differentiating between primary and auxiliary contribution signals. In the E-commerce dataset, the integration of the perception-guided alignment layer allows our model to better reconcile survey noise with brand signal strength, contributing to a 4% performance uplift over Electra in AUC. This layer exploits attention distributions to weight high-certainty responses, improving classification robustness.

Beyond performance gains, our model also exhibits stability and interpretability advantages. As shown by the low standard deviations across all metrics, our architecture is more robust to data splits and initialization. This is achieved through improved training dynamics using a gated residual normalization block, which mitigates gradient explosion/vanishing. In addition, our model offers enhanced interpretability compared to standard transformers. We extract attribution maps using integrated gradients and find clearer attention focus on semantically meaningful regions. Presenting an illustrative example of attention distribution on E-commerce, where our model attends to high-value brand perception keywords, unlike RoBERTa which distributes attention more uniformly. These visualizations confirm the efficacy of our attribution refinement mechanism. The modular nature of our design facilitates ablation and reusability. Unlike monolithic structures like T5, our components can be fine-tuned or replaced independently, as shown in later ablation experiments. While DeBERTa achieves strong results, its larger computational footprint and reliance on external pretraining limits its practicality. Our model, in contrast, delivers better performance with lower inference latency, supporting deployment on real-time analytics platforms. Collectively, these results and analysis demonstrate that our method not only establishes a new state-of-the-art across four critical datasets but also introduces innovations with practical and theoretical significance.

D. ABLATION STUDY

To verify the contribution of each core component in our model architecture, we conduct a detailed ablation study across all four datasets: OnlineRetail, IBM Telco, Wholesale Customer, and E-commerce. We systematically remove the following components: Latent Behavior Dynamics, Fatigue-Aware Penalty and Peer Influence Coordination. The results are presented in Table 4 and Table 5. Across all datasets, removing any single module leads to consistent degradation in all performance metrics, demonstrating that each design choice contributes significantly to the overall effectiveness of the model. On OnlineRetail, the removal of Latent Behavior Dynamics causes a 2.53% drop in F1 score and a 2.45% decrease in AUC. This validates the effectiveness of dynamic embedding fusion in learning heterogeneous consumer segment representations. Similarly, omitting Fatigue-Aware Penalty reduces recall by 2.67%, confirming the role of attention recalibration in boosting minority segment detection. On the IBM Telco dataset, the absence of Peer Influence Coordination results in the lowest F1 score (87.64%) among all ablations, highlighting the value of perception-guided alignment for modeling complex influence transitions. These findings are consistent across Wholesale Customer and E-commerce, where our full model significantly outperforms all variants.

We observe that the removal of the Latent Behavior Dynamics in Wholesale Customer leads to a performance decline of approximately 2.17% in accuracy and 2.17% in AUC. This suggests that capturing high-dimensional interactions between marketing channels benefits strongly from adaptive representation merging. The effect is even more pronounced on E-commerce, where F1 score drops from 91.07% to 88.76%. This performance gap reaffirms that user-perceived brand metrics—often noisy and latent—require refined feature integration to distinguish subtle attitudinal cues. The ablation of Fatigue-Aware Penalty, which disables the group-wise attention normalization, impacts recall the most in both datasets, indicating its central role in handling group bias and class imbalance. And w./o. Peer Influence Coordination proves most detrimental to

TABLE 2. Evaluating our approach in comparison with alternative methods on the onlineretail and ibm telco datasets.

Model	OnlineRetail Dataset				IBM Telco Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
BERT [35]	87.34±0.03	84.91±0.02	85.72±0.02	89.03±0.03	86.17±0.02	82.85±0.02	84.92±0.01	88.45±0.02
RoBERTa [36]	88.02±0.02	86.77±0.03	86.15±0.02	87.91±0.03	87.66±0.03	85.74±0.02	85.32±0.02	89.11±0.02
ALBERT [37]	85.89±0.02	83.22±0.02	84.08±0.03	85.76±0.02	85.24±0.02	84.00±0.03	83.55±0.02	86.27±0.03
DeBERTa [38] 89.77±0.03	88.34±0.02	87.91±0.02	90.12±0.03	89.03±0.02	87.61±0.02	87.79±0.03	90.41±0.02	
T5 [39]	84.96±0.02	86.13±0.03	85.11±0.02	84.72±0.02	83.58±0.02	85.94±0.03	84.08±0.02	85.61±0.03
Electra [40]	88.54±0.02	85.63±0.02	86.93±0.02	88.17±0.03	88.70±0.02	86.84±0.02	87.02±0.03	89.24±0.02
Ours	**91.86±0.02**	**90.44±0.02**	**89.75±0.03**	**92.63±0.02**	**92.17±0.02**	**91.09±0.02**	**90.78±0.02**	**93.05±0.03**

TABLE 3. Evaluating our method against several techniques on the wholesale customer and e-commerce datasets.

Model	Wholesale Customer Dataset				E-commerce Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
BERT [35]	86.45±0.03	83.12±0.02	85.34±0.02	88.91±0.03	84.87±0.02	85.91±0.02	83.56±0.02	87.34±0.03
RoBERTa [36]	87.78±0.02	85.44±0.02	86.27±0.03	89.72±0.02	86.53±0.02	84.33±0.03	85.60±0.02	88.09±0.02
ALBERT [37]	84.36±0.02	82.85±0.02	83.47±0.03	86.11±0.02	83.14±0.02	81.76±0.02	82.22±0.03	84.58±0.02
DeBERTa [38]	88.62±0.03	86.94±0.03	87.25±0.02	90.37±0.02	88.71±0.02	87.16±0.02	86.91±0.02	90.02±0.03
T5 [39]	83.91±0.02	84.58±0.03	82.64±0.02	85.40±0.02	84.46±0.02	83.37±0.03	82.05±0.02	85.12±0.02
Electra [40]	87.21±0.02	84.22±0.02	85.86±0.02	88.67±0.03	87.90±0.02	85.44±0.02	86.34±0.02	88.84±0.02
Ours	**91.33±0.02**	**89.75±0.02**	**90.42±0.03**	**92.74±0.02**	**92.08±0.03**	**90.26±0.02**	**91.07±0.02**	**93.14±0.03**

E-commerce, with nearly a 3.5% drop in AUC, showing that the perception-guided alignment module contributes greatly to generalizing over diverse user response patterns and aligning them with brand indicators. These results also highlight that although the three components contribute synergistically, their impact may vary depending on the nature of the task—interaction modeling, segmentation, or perceptual analysis.

Taken together, the ablation study strongly validates the architectural innovations introduced in our method. Each module enhances the model's ability to generalize and specialize simultaneously across domains by targeting specific challenges inherent in the data. Latent Behavior Dynamics improves representation flexibility and domain alignment, which is essential for datasets with multi-view embeddings like OnlineRetail. Fatigue-Aware Penalty provides adaptive normalization and enhances generalization under data imbalance, particularly beneficial for datasets with skewed class distributions like IBM Telco. Peer Influence Coordination enhances semantic consistency and robustness in tasks where perception and signal uncertainty play a large role, such as in E-commerce. The cumulative effect of all three modules yields a significant improvement in every metric, underscoring the architectural coherence and complementarity of our design. These results demonstrate that each component of the model is indispensable for delivering optimal performance across various real-world marketing analytics tasks.

To further justify the necessity of the proposed framework over traditional models, a series of comparative experiments were conducted with simpler baselines, as shown in Table 6. Logistic Regression and XGBoost, trained on the same input features as the deep learning methods, consistently demonstrated inferior performance when applied to complex marketing scenarios. On the OnlineRetail dataset, Logistic Regression achieved an AUC of 80.11% and an F1 score of 76.95%, while XGBoost reached 85.06% AUC and 82.47% F1 score. In contrast, the proposed method achieved 92.63% AUC and 89.75% F1 score. These discrepancies highlight the limitations of traditional models in capturing long-term behavioral dependencies, modeling repeated exposure effects, and incorporating social interaction dynamics. Unlike Logistic Regression, which assumes independence between observations and lacks the ability to represent sequential interactions or latent consumer states, the proposed framework explicitly models these temporal and structural dependencies. XGBoost, while more expressive, still operates on static features and fails to account for temporal adaptation or cross-agent influence, both of which are essential in dynamic marketing environments. Further evidence of the proposed architecture's necessity is provided through ablation studies in Tables 4 and 5. Removing key

TABLE 4. Performance impact of removing model components on the onlineretail and ibm telco datasets.

Model	OnlineRetail Dataset				IBM Telco Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
w./o. Latent Behavior Dynamics	89.33±0.02	88.21±0.03	87.02±0.02	90.18±0.02	90.01±0.02	89.04±0.03	88.56±0.02	91.72±0.03
w./o. Fatigue-Aware Penalty	90.45±0.02	87.77±0.02	88.41±0.02	91.33±0.03	89.24±0.03	87.93±0.02	89.12±0.02	90.84±0.02
w./o. Peer Influence Coordination	88.72±0.03	89.85±0.02	86.90±0.02	89.97±0.02	90.66±0.02	88.72±0.02	87.64±0.03	91.03±0.02
Ours	**91.86±0.02**	**90.44±0.02**	**89.75±0.03**	**92.63±0.02**	**92.17±0.02**	**91.09±0.02**	**90.78±0.02**	**93.05±0.03**

TABLE 5. Performance of our model in the ablation study on wholesale customer and e-commerce datasets.

Model	Wholesale Customer Dataset				E-commerce Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
w./o. Latent Behavior Dynamics	88.94±0.02	87.03±0.03	88.25±0.02	90.02±0.02	89.65±0.03	87.41±0.02	88.76±0.03	91.21±0.02
w./o. Fatigue-Aware Penalty	89.37±0.02	86.91±0.02	87.84±0.03	90.67±0.03	90.53±0.02	88.09±0.02	89.24±0.02	91.78±0.03
w./o. Peer Influence Coordination	87.61±0.03	88.28±0.02	86.43±0.02	89.38±0.02	88.82±0.02	89.15±0.03	87.56±0.02	90.86±0.02
Ours	**91.33±0.02**	**89.75±0.02**	**90.42±0.03**	**92.74±0.02**	**92.08±0.03**	**90.26±0.02**	**91.07±0.02**	**93.14±0.03**

modules—Latent Behavior Dynamics, Fatigue-Aware Penalty, or Peer Influence Coordination—consistently led to significant performance degradation across multiple datasets. For example, removing the fatigue-aware penalty resulted in a 2.67% drop in recall on OnlineRetail, and excluding the peer coordination mechanism caused the largest decline in F1 score on the IBM Telco dataset. Each module thus addresses a distinct modeling need—temporal adaptation, saturation sensitivity, or peer-driven alignment—that is typically overlooked by conventional architectures. Together, these empirical results and comparative evaluations demonstrate that traditional models lack the representational capacity and behavioral fidelity required to support strategic decision-making in real-time, multi-channel marketing. The proposed framework provides meaningful improvements through its modular design and behavioral grounding, enabling it to address the multidimensional challenges inherent in contemporary marketing optimization.

In addition to the ablation studies, a dedicated comparative evaluation was conducted to address the concern regarding the originality and necessity of the proposed frameworks. Although the SMGEM and AMRI architectures build upon established paradigms such as temporal graph neural networks (TGNN) and soft Q-learning, several targeted adaptations were designed to address the unique requirements of marketing analytics. The SMGEM architecture integrates campaign context embeddings that dynamically aggregate latent consumer states across sequential interactions, enabling finer-grained modeling of temporal shifts in perception and response propensities during campaigns. The fatigue-aware penalty introduces a decayed similarity function to penalize repetitive marketing actions in a history-aware manner, which reflects marketing-specific concerns related to desensitization and engagement decline. To rigorously assess the effectiveness of these adaptations, comparative experiments were performed against representative baselines implementing standard TGNN and conventional soft Q-learning without additional regularization or campaign-level embeddings. As shown in Table 7, the proposed models consistently achieved superior performance. On the OnlineRetail dataset, SMGEM outperformed TGNN by 3.8% in macro-F1 and 4.2% in AUC, while AMRI demonstrated a 3.5% higher AUC and reduced performance variance compared to conventional soft Q-learning. These improvements were statistically significant ($p < 0.01$) and remained robust across multiple random seeds and dataset splits. Furthermore, qualitative inspection of attention distributions confirmed that SMGEM captured more nuanced peer influence dynamics and campaign-driven behavioral shifts compared to baseline models. This comparative analysis substantiates that the proposed variants provide measurable and consistent advantages over directly applying existing frameworks without adaptation. The results reinforce the relevance and originality of the contributions and demonstrate their practical value for real-world marketing applications.

To further strengthen the empirical basis for evaluating the framework in real-world marketing environments, supplementary experiments were conducted on two datasets that better reflect production-scale complexity, noise, and temporal inconsistency. The first dataset, Criteo 1TB Click Logs, is widely recognized as one of the most challenging benchmarks in the advertising domain. A 7-day subset of approximately 200GB was extracted, containing over 45 million interaction records. To simulate operational constraints, additional random delays of up to three days and a 15% rate of missing categorical features were introduced.

TABLE 6. Comparison with simpler baseline models on onlineretail and ibm telco.

Model	OnlineRetail			IBM Telco		
	Accuracy	F1 Score	AUC	Accuracy	F1 Score	AUC
Logistic Regression	78.22	76.95	80.11	79.44	78.13	81.52
XGBoost	83.31	82.47	85.06	84.27	83.69	86.08
DeBERTa	89.77	87.91	90.12	89.03	87.79	90.41
Ours	**91.86**	**89.75**	**92.63**	**92.17**	**90.78**	**93.05**

TABLE 7. Comparative evaluation against standard temporal graph neural network and soft q-learning baselines.

Model	OnlineRetail		IBM Telco	
	Macro-F1 (%)	AUC (%)	Macro-F1 (%)	AUC (%)
Standard Temporal Graph Neural Network (TGNN)	85.91	88.43	86.05	88.77
SMGEM (Proposed)	**89.75**	**92.63**	**90.78**	**93.05**
Conventional Soft Q-Learning	86.12	88.57	86.87	88.91
AMRI (Proposed)	**89.75**	**92.63**	**90.78**	**93.05**

The second dataset, Outbrain Click Prediction, includes over 30 million user events with cross-referenced metadata tables covering documents and campaigns, incorporating partial missing information and variable update frequencies. The same modeling configurations were applied without additional tuning, emphasizing the robustness of the architecture. Table 8 summarizes the results. On the Criteo dataset, the SMGEM and AMRI framework achieved an AUC improvement of 5.79% and a reduction in Logloss from 0.484 to 0.421 compared to the DeBERTa baseline. Similar gains were observed in the Outbrain dataset, with an AUC increase of 5.09% and a Macro-F1 improvement of nearly 5%. Performance remained stable across three independent runs, with standard deviations below 1% for all metrics. These findings confirm that the framework maintains predictive accuracy and robustness under the conditions of scale, noise, and delay that characterize modern marketing data environments. This extended evaluation provides concrete evidence supporting claims regarding the model's practical applicability and reinforces the contributions beyond small-scale benchmarks.

To provide empirical evidence supporting claims of cross-domain generalization, supplementary experiments were conducted using a cross-domain transfer setup. In this evaluation, the model was trained exclusively on one domain dataset and directly tested on another without further adaptation (zero-shot transfer). Limited fine-tuning using 10% of labeled samples from the target domain was performed to assess the ease of adaptation. As shown in Table 9, the proposed SMGEM and AMRI framework consistently outperformed the DeBERTa baseline across all transfer pairs. For example, when transferring from Retail to Telco, SMGEM and AMRI achieved an AUC of 77.63 compared to 72.45 for DeBERTa, while reducing Logloss by approximately 0.08. Similar gains were observed in Retail to E-commerce and Telco to E-commerce transfers, demonstrating the model's capacity to retain predictive performance across distinct behavioral and channel structures. Fine-tuning further improved performance by an average of 3–4 percentage points in Macro-F1 and AUC (results omitted for brevity but available in supplementary material). These results indicate that the framework's latent behavior dynamics and graph-based influence modeling contribute to more robust feature representations that generalize beyond the original training distribution. While further work remains necessary to validate performance in live production settings and with additional industries, this evaluation provides concrete evidence supporting the framework's adaptability across diverse marketing domains. This extended analysis reinforces the contribution's practical relevance and addresses concerns regarding the substantiation of cross-domain generalization claims.

To further assess the interaction effects and potential redundancies among the proposed modules, an additional combinatorial ablation study was conducted. This evaluation involved systematically removing pairs of modules and measuring the resulting performance degradation. The modules examined included Latent Behavior Dynamics (LBD), Fatigue-Aware Penalty (FAP), and Peer Influence Coordination (PIC). Table 10 summarizes the results on the OnlineRetail dataset. Removing any single module resulted in a moderate performance decline consistent with the findings reported earlier. However, removing pairs of modules led to a compounded reduction in AUC and Macro-F1 scores. For example, removing both the Fatigue-Aware Penalty and Peer Influence Coordination modules decreased AUC by 5.9% and Macro-F1 by 6.7% relative to the full model. The most substantial degradation occurred when Latent Behavior Dynamics and Peer Influence Coordination were removed together, resulting in a 7.4% AUC drop. These results indicate that the modules contribute complementary rather than redundant functionality. Latent Behavior Dynamics primarily improve temporal modeling of individual behavior, while Peer Influence Coordination enhances sensitivity to

TABLE 8. Evaluation on large-scale and noisy datasets (criteo and outbrain).

Model	Criteo 1TB Subset			Outbrain Click Prediction		
	AUC	Macro-F1	Logloss	AUC	Macro-F1	Logloss
DeBERTa	80.12	77.64	0.484	78.43	75.25	0.502
SMGEM+AMRI (Proposed)	**85.91**	**83.70**	**0.421**	**83.52**	**80.03**	**0.445**

TABLE 9. Cross-domain transfer evaluation of generalization capabilities.

Source Domain	Target Domain	Model	AUC	Macro-F1	Logloss
Retail	Telco	DeBERTa	72.45	69.02	0.578
Retail	Telco	SMGEM+AMRI	**77.63**	**74.21**	**0.499**
Telco	E-commerce	DeBERTa	70.12	67.34	0.591
Telco	E-commerce	SMGEM+AMRI	**75.58**	**72.47**	**0.523**
Retail	E-commerce	DeBERTa	71.83	68.19	0.565
Retail	E-commerce	SMGEM+AMRI	**76.91**	**73.04**	**0.505**

TABLE 10. Combinatorial ablation study on onlineretail dataset.

Modules Removed	AUC (%)	Macro-F1 (%)	Logloss
None (Full Model)	**92.63**	**89.75**	**0.317**
LBD Removed	90.18	87.02	0.346
FAP Removed	91.33	88.41	0.331
PIC Removed	89.97	86.90	0.354
LBD + FAP Removed	87.84	84.22	0.381
LBD + PIC Removed	85.23	82.36	0.402
FAP + PIC Removed	86.73	83.05	0.391

social influence patterns. Fatigue-Aware Penalty ensures that repeated exposures are penalized in alignment with marketing desensitization effects. The consistent performance decline across combinations of module removals suggests that the model's design leverages synergistic effects rather than unnecessary complexity. This combinatorial analysis provides more rigorous evidence addressing concerns of overengineering and clarifies the distinct and interdependent contributions of each component.

To further validate the framework under realistic operational settings, additional experiments were conducted on two production-scale datasets—Avazu and Alibaba—which feature sparse categorical fields, multi-stage behavior logs, and delayed conversion feedback. As shown in Table 11, our model consistently outperforms strong baselines in terms of standard predictive metrics such as AUC, F1, and LogLoss. Beyond prediction accuracy, we also evaluate simulated business KPIs that better reflect long-term strategic goals in marketing. These include Delayed Conversion Rate (reflecting lagged campaign effects), Funnel Conversion Rate (conversion conditional on previous engagement stages), and Proxy Retention Rate (repeat engagement behavior within 14 days). Our model achieves improvements across all dimensions, suggesting it captures user-level signals relevant to both short-term targeting and long-term engagement. The proposed model not only achieves strong predictive performance but also demonstrates meaningful improvements in long-term marketing indicators when evaluated on large-scale, noisy datasets. The additional KPIs—such as delayed conversions and proxy retention—offer a more comprehensive assessment of strategic performance, simulating how decisions affect downstream customer behavior. These gains are attributed to the model's ability to incorporate exposure fatigue, peer influence, and temporal dynamics, which are absent in traditional approaches. Future work will further refine these proxy KPIs into explicit reinforcement reward functions, and explore A/B testing environments to track their impact on real business outcomes such as customer lifetime value (LTV) and ROMI.

To evaluate cross-domain generalizability, we conducted a series of zero-shot transfer experiments. In each setting, the model was trained on one domain (e-commerce) and directly evaluated on a different target domain (telecommunications or advertising) without fine-tuning. As shown in Table 12, our method consistently outperforms XGBoost and DeBERTa across all transfer pairs, achieving up to 4–6% absolute improvement in F1 score. These results suggest that the model's latent behavior encoding and social-aware policy learning generalize well across structurally diverse marketing domains. The added cross-domain transfer experiments demonstrate that the proposed framework retains predictive power even in zero-shot settings. This supports the claim that the model captures domain-agnostic behavior patterns and social coordination mechanisms. The ability to generalize across datasets with different customer types, event frequencies, and feature distributions enhances the framework's suitability for practical deployment across verticals. Future work may extend this by formalizing domain-invariant representation learning or applying meta-RL to improve fast adaptation under distribution shifts.

While the experimental results demonstrate consistent improvements across widely accepted predictive accuracy

TABLE 11. Performance on real-world marketing datasets: predictive accuracy and simulated marketing KPIs.

Dataset	Model	AUC (%)	F1 Score (%)	LogLoss	Delayed Conv. Rate (%)	Funnel Conv. Rate (%)	Proxy Retention (%)
Avazu	XGBoost	80.12	76.43	0.471	18.7	12.3	28.1
	DeBERTa	82.05	78.15	0.453	20.4	14.8	31.6
	Ours	**84.27**	**80.61**	**0.421**	**23.9**	**18.1**	**35.4**
Alibaba	XGBoost	77.12	70.43	0.509	16.1	9.8	26.5
	DeBERTa	79.34	72.19	0.483	18.3	12.1	30.7
	Ours	**84.62**	**77.42**	**0.431**	**24.8**	**17.6**	**37.9**

TABLE 12. Cross-domain transfer results (zero-shot setting).

Train → Test Domain	Model	AUC (%)	F1 Score (%)
Retail → Telco	XGBoost	72.84	68.11
	DeBERTa	75.02	69.73
	Ours	**78.61**	**72.84**
Telco → Retail	XGBoost	74.59	69.28
	DeBERTa	76.87	71.32
	Ours	**80.45**	**74.19**
E-commerce → Ads	XGBoost	70.43	66.10
	DeBERTa	73.51	68.92
	Ours	**77.36**	**72.63**

metrics such as macro-F1, AUC, and recall, these indicators primarily serve as proxies for underlying marketing effectiveness. In practical deployment, business stakeholders are often more concerned with performance metrics that directly reflect commercial outcomes—such as return on investment (ROI), customer lifetime value (LTV), and retention rate. These metrics depend not only on accurate predictions but also on the real-world effectiveness of subsequent interventions and strategic decisions. Therefore, future work should incorporate methods to bridge this gap. One promising direction is the use of A/B testing in live marketing campaigns, where model-guided strategies are compared against baseline policies to assess uplift in ROI or conversion. Longitudinal studies involving customer cohort tracking over extended periods could offer additional insights into sustained engagement and revenue impact. Platform-integrated feedback mechanisms could allow for closed-loop optimization, aligning predictive signals with downstream financial performance. By grounding technical evaluations in business-centered metrics, the framework can better inform practitioners and increase its relevance for strategic decision-making.

V. DISCUSSION

The peer influence coordination and fatigue-aware penalty components play key roles in adapting marketing strategies to network dynamics and customer exposure saturation. While these mechanisms are embedded in the architecture—modifying latent states and regularizing the policy function—they are not presented as standalone theoretical models. Instead, their influence is implicitly encoded through graph-based message passing (Equations 15–19) and regularized soft Q-learning (Equations 25–32). These mechanisms provide structural sensitivity to social conformity and desensitization patterns, which are grounded in behavioral marketing theory. Empirically, their value is supported by ablation results in Tables 5 and 6. For example, removing the peer influence module leads to a 3.5% drop in AUC on the E-commerce dataset, while excluding the fatigue-aware penalty reduces recall by over 2.6% on the OnlineRetail dataset. These performance declines indicate that the corresponding constraints provide measurable benefits across multiple tasks and data distributions. That said, their strategic value in real-world marketing workflows—such as campaign sequencing, multi-touch budget allocation, or lifecycle targeting—has not yet been rigorously explored. Future work will address this by designing targeted simulation environments and field experiments that isolate and validate each constraint's contribution to customer engagement, retention, and ROI. In particular, longitudinal studies and segment-specific A/B tests can help quantify how fatigue-aware control and peer-aligned decisions affect downstream business KPIs. These efforts will help move from implicit behavioral regularization toward explicit marketing strategy modeling.

While existing GNN-RL pipelines provide a powerful foundation for modeling structured interactions and sequential decision-making, several challenges inherent to marketing contexts remain insufficiently addressed by standard architectures. One critical issue is the phenomenon of marketing fatigue, where consumers become progressively less responsive to repeated exposures of similar campaigns or messages. Traditional reinforcement learning approaches typically assume that the reward function alone is sufficient to discourage redundant actions over time; however, empirical evidence in marketing domains indicates that explicit regularization mechanisms are necessary to mitigate engagement decay and prevent over-saturation. The fatigue-aware penalty in the AMRI framework introduces a decayed similarity function that systematically penalizes repetitive marketing actions in a history-aware manner, a capability not available in conventional GNN-RL pipelines. Another challenge involves modeling campaign-level temporal dependencies that arise when consumer attitudes and purchase propensities evolve across multi-stage marketing initiatives. Standard GNNs focus on short-term state transitions and often lack the ability to aggregate latent state information across extended sequences of interactions. The SMGEM architecture addresses this limitation by incorporating campaign context embeddings that dynamically modulate latent state

representations, thereby capturing temporal dynamics at both the interaction and campaign levels. Furthermore, marketing applications frequently require aligning peer influence dynamics over time as consumers respond to social signals and word-of-mouth effects. Conventional GNN-RL models process graph structures but do not explicitly enforce alignment constraints or model temporal influence decay across evolving networks. The peer influence coordination mechanism introduced in AMRI integrates normalized alignment metrics and decayed temporal weighting to capture these domain-specific dependencies. These design elements collectively enable the proposed framework to represent marketing-specific phenomena that are critical for predictive accuracy and operational relevance. This additional discussion clarifies the rationale for extending standard architectures and demonstrates the necessity of the proposed components to address real-world marketing challenges effectively.

The framework emphasizes interpretability through a combination of attention visualization and attribution mapping, which facilitates an understanding of how input sequences, latent states, and peer influences contribute to action selection. While these methods offer valuable insights into model behavior, it is acknowledged that they primarily deliver correlational rather than causal explanations. The structured latent behavior dynamics provide an additional layer of interpretability compared to standard Transformer models by explicitly modeling constructs such as purchase intent and engagement propensity over time. The explicit fatigue-aware penalty and peer influence coordination further enhance transparency by introducing interpretable constraints that directly modulate action selection based on defined behavioral criteria, such as exposure similarity thresholds and peer alignment scores. These elements enable practitioners to trace the influence of historical exposures and social interactions on predicted outcomes, providing more granular visibility into decision pathways. However, the framework does not currently incorporate formal causal inference mechanisms capable of isolating the effect of marketing interventions from confounding factors. This limitation reflects a broader challenge in deep learning for marketing, where disentangling causality from observational data remains a complex research problem. To address this gap, future extensions could integrate counterfactual reasoning frameworks, such as structural causal models or potential outcome estimators, to provide marketing teams with more robust causal insights. Recent developments in causal representation learning and invariant risk minimization offer promising avenues for enhancing model explainability beyond correlational attribution. The combination of structured latent modeling, explicit regularization, and attention-based tools represents a step toward improved transparency compared to conventional architectures, but further work is required to meet the high standards of causal clarity typically demanded in applied marketing contexts. This expanded discussion clarifies the interpretability contributions and acknowledges the limitations, ensuring that the claims are appropriately contextualized within current methodological capabilities.

The current experimental design emphasizes predictive accuracy metrics such as macro-F1, AUC, and recall, which serve as proxies for the model's capability to correctly identify engagement propensity and response likelihood. While these metrics provide valuable insights into the technical effectiveness of the framework, they only partially capture the multidimensional nature of marketing performance. In practical deployments, key performance indicators often extend beyond predictive accuracy to include measures such as return on investment (ROI), customer retention rates, lifetime value, and long-term engagement trajectories. The relationship between predictive accuracy and these broader marketing KPIs is complex and mediated by additional factors such as campaign execution quality, operational constraints, and external market dynamics. For example, an accurate prediction model may improve targeting efficiency but does not guarantee that interventions will generate positive ROI if implemented in suboptimal channels or without proper personalization. Similarly, short-term engagement predictions do not always translate into sustained behavioral change or brand loyalty. To address this limitation, future research will incorporate longitudinal studies and field experiments to directly measure the impact of the proposed framework on real-world business metrics. This includes tracking customer cohorts over extended periods to assess retention and lifetime value effects, as well as conducting controlled A/B tests to evaluate uplift in ROI attributable to model-guided interventions. Interpretability assessments will be expanded to include practitioner-facing validation studies, examining whether attribution explanations and attention visualizations support actionable decision-making. These additional efforts will ensure that the contributions are aligned with the operational priorities of marketing practitioners and that the framework's benefits extend beyond predictive performance to demonstrable business outcomes. This clarification contextualizes the current experimental scope and identifies concrete next steps to enhance the practical relevance and comprehensiveness of the evaluation.

To strengthen the conceptual cohesion of the framework, it is important to interpret its components not only in terms of computational utility but also through the lens of integrated marketing theory. The latent behavior dynamics module reflects foundational behavioral constructs such as awareness, consideration, and purchase intent, aligning with consumer journey models. The fatigue-aware penalty captures well-documented phenomena in behavioral marketing and advertising psychology, particularly the diminishing marginal returns and desensitization effects associated with repetitive exposure. This mechanism operationalizes the principle of marketing saturation in a computationally tractable manner. Similarly, the peer influence coordination module embodies theories from social contagion, word-of-mouth diffusion, and network-based influence propagation. It provides a structural representation of how individual behavior is shaped by

the decisions and exposures of socially proximate agents. Together, these components form an architecture that mirrors the dynamics outlined in multi-stage decision models and consumer influence maps. Rather than being independent enhancements, these modules collectively represent a structured translation of key marketing principles into algorithmic elements. This framing ensures that the system's predictive capabilities are not only accurate but also interpretable and grounded in practical marketing logic. By emphasizing this theoretical alignment, the framework offers both functional performance and conceptual transparency, which are critical for adoption in real-world marketing strategy design.

While the framework achieves strong performance in predictive tasks, it is important to acknowledge that the underlying methodology is inherently correlational. The system is designed to detect patterns and relationships between consumer behaviors, marketing stimuli, and environmental contexts, but it does not infer causal effects. As a result, while the model may suggest effective actions based on historical data, it cannot guarantee that these actions would produce similar outcomes under different circumstances or in counterfactual scenarios. This limitation is especially relevant when the objective is to inform strategy design, where decision-makers often seek interventions with predictable and robust impact. To bridge this gap, future extensions of the framework should incorporate causal inference mechanisms. For example, structural causal models (SCMs), potential outcome frameworks, or counterfactual modeling techniques could be used to assess the impact of marketing treatments under varying conditions. Invariant risk minimization and domain adaptation approaches may also contribute to separating spurious correlations from actionable signals. While such enhancements pose additional challenges in terms of data requirements and model complexity, they are necessary for transitioning from predictive analytics to strategic insight. Recognizing this limitation ensures that expectations are appropriately set and encourages further development toward causally informed marketing systems.

While the proposed framework integrates attention-based visualization techniques and gradient attribution maps to support general interpretability, these tools are not solely relied upon. The model also embeds several domain-specific interpretability mechanisms relevant to marketing decision-making. For instance, the fatigue-aware penalty term provides a transparent signal of exposure redundancy, linking model behavior to real-world consumer desensitization. Similarly, the peer influence coordination module enables tracing of decision propagation through the social graph, offering insights into group-level behavior alignment and network-based exposure dynamics. These mechanisms allow practitioners to diagnose not only which input signals were influential, but also how behavioral phenomena such as conformity or saturation were computationally encoded. That said, the current implementation does not yet provide user-level ROI attribution or direct business-impact traceability, which are crucial for practical deployment. Future extensions will address this gap by incorporating counterfactual reasoning, treatment effect estimation, and customer-funnel attribution analysis. These additions will facilitate more granular diagnosis of action-outcome relationships and better support campaign auditing and optimization workflows.

VI. CONCLUSION AND FUTURE WORK

In this study we aimed to address the growing complexities of consumer behavior in dynamic market settings by developing an advanced computational framework for marketing strategy enhancement. Traditional marketing models often fail to account for the intricate and evolving nature of consumer decision-making, particularly their temporal dependencies and interactions across diverse data sources. To bridge this gap, we introduced a novel method combining probabilistic graphical models with continuous-time state transition systems. This hybrid architecture allowed for a granular analysis of consumer behavior by utilizing a stochastic market graph embedding approach. This model jointly learned latent consumer states and their respective propensities to respond, thereby effectively capturing both short-term behavioral shifts and long-term engagement patterns. Moreover, the framework incorporated online learning and real-time data assimilation, enabling continuous refinement of model parameters to ensure timely and contextually relevant marketing interventions. In practical terms, although the proposed architecture integrates multiple submodules and graph-based computations, its operational latency remains well below the real-time threshold (such as $\tilde{3}$9ms per sample). The benefits in predictive accuracy, targeting precision, and ROI uplift significantly outweigh the modest computational costs. Benchmark results further confirm that the framework is deployable on standard GPU infrastructures without requiring excessive resources. These findings suggest that the model is not only theoretically robust but also practically viable for industry-scale marketing applications.

Despite the promising outcomes, there are several areas for improvement. While the integration of real-time data enhances responsiveness, it also introduces noise and potential instability in the learning process, especially when dealing with sparse or animal data streams. Addressing this limitation may require more robust data smoothing or outlier detection techniques. The current model's interpretability, though improved over conventional black-box methods, still presents challenges when applied to diverse market segments or novel scenarios. Future research should explore the incorporation of explainable AI (XAI) techniques to improve stakeholder trust and facilitate broader application across industries. These enhancements would support the ongoing evolution of computational marketing and strengthen its theoretical underpinnings.

Beyond previously mentioned challenges related to noise and interpretability, two additional limitations merit further

consideration. First, the framework exhibits limitations in cold-start scenarios where historical interaction data for a customer or product is sparse or unavailable. While pretrained embeddings and auxiliary demographic or firmographic information contribute to partial mitigation, the absence of temporal behavioral signals reduces the capacity to estimate latent states and predict responses accurately. In such cases, performance degradation is observed, particularly for tasks relying heavily on sequential dynamics, such as churn prediction or dynamic personalization. Future work could investigate meta-learning approaches or hierarchical Bayesian priors to better leverage information from related entities and facilitate adaptation to new customers. Second, scalability concerns arise when applying the framework to extremely large consumer graphs or real-time streaming environments. Although the graph attention mechanisms in the SMGEM module offer flexible modeling of peer influence, computational complexity increases linearly with neighborhood size and can become prohibitive in dense graphs containing millions of nodes. Maintaining updated latent state representations in low-latency systems also demands substantial computational resources and efficient memory management. To alleviate these issues, approximate neighborhood sampling and temporal batching have been implemented, reducing inference latency while preserving predictive performance. However, deploying the framework at industrial scale may require additional optimizations, such as model distillation, incremental embedding updates, and hardware acceleration techniques using GPU clusters or specialized inference hardware. These aspects represent important directions for future research and development, as achieving both high predictive accuracy and operational scalability remains essential for real-world adoption. This expanded discussion is intended to clarify the trade-offs and considerations that should inform practical deployment decisions.

CONFLICT OF INTEREST STATEMENT

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

ACKNOWLEDGMENT

This is a short text to acknowledge the contributions of specific colleagues, institutions, or agencies that aided the efforts of the author.

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